

The Rise of Alternative Pathways

In the Evolving Job Market

by

Isobel Bartels, Braeden Milar, Cameron Da Cunha

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Abstract

The traditional belief that a bachelor's degree is the only path to career success has long shaped educational expectations, despite rising evidence of its limitations. Research before 2020 highlighted issues such as low degree completion rates, high student debt burdens, and the emergence of vocational and skills-based alternatives as credible routes to employment. The COVID-19 pandemic dramatically accelerated these trends, exposing systemic weaknesses in higher education and forcing both students and employers to rethink what constitutes value in education. Today, alternative pathways like boot camps, certifications, and career-connected learning programs are gaining legitimacy, offering students quicker and more affordable routes into the workforce. Efforts are already underway to adapt to this shifting educational landscape, including proposals to expand credentials, redefine student support systems, and move toward competency-based hiring practices. However, substantial gaps remain in research and policy. Three key areas that require further attention are standardizing definitions for nontraditional students to ensure better institutional support, tracking long-term career outcomes across diverse educational paths, and reforming financial aid and advising structures to align with a broader set of educational options.

The Rise of Alternative Pathways In the Evolving Job Market

Today, more than 60% of students question whether pursuing a college degree is worth the cost (Carnevale, Cheah, & Hanson, 2021). Rising tuition prices, soaring student debt, and the evolving demands of the modern workforce have shifted the longstanding assumption that a bachelor's degree is the surest path to success. Alternative education pathways such as vocational programs, boot camps, certificates, and online learning are rapidly gaining visibility and credibility.

This paper seeks to answer the central question: How are alternative education and career pathways reshaping access to employment and career advancement in today's evolving job market?

Understanding this shift is crucial because students are already facing unprecedented decisions about how best to invest in their futures. In a world where skills often matter more than prestige, students must navigate new opportunities while avoiding outdated assumptions that could lead to debt, dissatisfaction, or instability. Our goal is to illuminate the changes underway, why they matter, and how students can make more strategic, empowered choices in this transformed landscape.

II. What We Knew Then: Pre-2020 Research & Insights

In the early 2000s, the expectation was that everyone would earn a bachelor's degree. Multiple sources show that, at the time, open admission policies were prevalent, making it extremely easy for anyone to get into college. Open admission policies are where universities accept anyone who meets their minimum requirements, without considering the student as a whole. This initiative was initially implemented to increase college attendance. Eventually, these practices came to be criticized by scholars for not being a good fit for most students. Many students enrolled in college with the expectation of earning a BA, yet fewer than half of high school seniors planning to do so succeed, with completion rates dropping below 20% for low-achieving students (Rosenbaum et al., 2010). This discrepancy shows the

pitfalls of promoting a one-size-fits-all approach to postsecondary education. It's in the 2010s that we begin to see a shift in the perspectives of Bachelor's degrees, to something less essential than what they once were.

With the overall sentiment shifting to favoring alternative education pathways, more research is being conducted on the subject. We see a 2013 study, “Adolescent Precursors of Pathways from School to Work,” that examines new school-to-work (STW) trends. The author Vuolo identified four key trajectories:

1. Obtaining a bachelor's degree and transitioning into a career (“Bachelor's into Career”)
2. Completing an associate or vocational degree and entering a career (“Associate/Voc-Tech into Career”)
3. Attending college without graduating and struggling to find stable employment (“Some College Floundering”)
4. Not pursuing higher education at all, leading to continued instability in the workforce (“High School Floundering”).

These categories show the depth of understanding and options for alternative career pathways at the time (2013). The STW trajectories were developed using longitudinal data from the Youth Development Study, and only one of the four is a genuine alternative to the standard bachelor's degree. Also, the vocabulary used, describing not pursuing higher education as “leading to continued instability in the workforce,” shows the connotations associated with not pursuing higher education in the early 2010s.

However, the same study supports students who fall into the “Associate/Voc-Tech into Career” category. The study found that youth who pursue associate or vocational degrees often have greater success establishing careers than those who attempt but fail to complete a four-year college program. This challenged the dominant train of thought that college is the only legitimate route to career success. People in the “Some College Floundering” group mostly invested in education without getting the full benefits of

the degree, leaving them worse off than those who pursued vocational training. Even in the beginning inquiries into alternative education paths, we see data that supports these growing options as being beneficial.

Carnevale, Rose, and Hanson's (2012) report also challenged the common belief at the time that a bachelor's degree was always the best or only path to career success. They showed that earning certificates or associate degrees in high-demand fields, such as computer science, healthcare, or engineering technology, often led to higher wages than obtaining a four-year degree in fields like education or social work. They also pointed out that shorter, skills-based programs helped students enter the workforce faster and avoid large amounts of debt. Back then, this was an early indicator of the trends we see today, with opinions on alternative education improving. It suggested that students needed to be more strategic, weigh the costs and outcomes of different education paths rather than assuming a bachelor's degree for every professional career.

While Carnevale, Rose, and Hanson (2012) showed that alternative credentials could offer substantial short-term financial benefits, other research from the same time period raised essential concerns about long-term outcomes. Hanushek, Schwerdt, Woessmann, and Zhang (2017) studied how vocational education compared to general education over a whole career. They found that even though vocational training helped individuals secure jobs more quickly in the early stages of their careers, the benefits often diminished as workers aged. Over time, those with broader general education backgrounds tended to have better employment prospects. This showed that while job-specific skills were valuable for entering the workforce, they did not always provide the adaptability needed for long-term career growth in evolving job markets. From a student's perspective back then, this research highlighted the need to focus not only on getting hired quickly but also on achieving long-term career stability and growth. It reminded students that choosing an education path required balancing short-term wins with long-term flexibility.

As concerns regarding the limits of traditional education pathways grew, new models gained popularity. Hoffman and Schwartz (2016) introduced the career pathways movement, which was intended to combine academic study with real work experience. They described how career pathways were written to offer students more connections between school learning and what they would be doing in the workplace. Instead of taking school and work as two sequential stages, the model advocated for simultaneous programs that blended technical training, classroom instruction, and on-the-job training. Back then, the model was regarded as an ambitious approach to affording greater access to safe occupational careers for a larger number of students, with special consideration towards students who traditionally had less opportunity to benefit.

In addition to the career pathways initiative, international studies were also beginning to emphasize the importance of more flexible and market-responsive education systems. In a 2010 report on the most advanced school systems globally, McKinsey analyzed how they had improved so significantly and found that one key area was their ability to respond rapidly to new economic demands. Education models that accommodated what employers required and offered multiple entry points based on student needs were more effective than rigid, one-size-fits-all models. This research supported the idea that education needed to be more flexible if it was going to set students up for real success, especially as businesses began to change at a faster pace.

Building on this idea, McKinsey's 2012 report, *Education to Employment*, tackled the issues of the traditional pipeline from school to work head-on. They identified three significant gaps that were harming students' futures: a skills gap, an expectation gap, and an information gap. The majority of the students were finishing up without the employers' hoped-for practical experience, amused themselves with fanciful illusions about their employment prospects, and had no useful guidance on the courses of action that would lead them to stable jobs. This research showed that even as more paths to education became accessible, the basic connection between school and work remained highly loose to most young

people. It led the conversation to rethink not just what students were studying, but how the entire system led them to real careers.

Ultimately, Deloitte's 2019 report showed how the job market itself was changing, making non-traditional pathways even more essential. They cited the rise of the gig economy and new models of employment, in which short-term contracts, freelancing, and flexible working arrangements have grown. This shift in requirements meant that prospective employees no longer needed just a degree, but a toolkit of adaptable skills and experiences to navigate an increasingly precarious career environment. This study at the time indicated that education systems needed to be sped up further, preparing students not just for a single safe job, but for a career lifetime of changing work opportunities.

III. Progress to Date: Contemporary Research & Field Developments

Since 2020, the conversation surrounding college and its alternatives has shifted from a place of speculation to transformation and action within education. The pandemic pushed forward the trends and growing concerns that had been building and pushing both students and employers to rethink the pathways that lead to career success. Now, today we can see that the research confirms what earlier studies were starting to show: alternative credentials are not just practical; they are in many scenarios the best option for many learners. Post-2020 data reveals a shift in the ROI of postsecondary education. A 2021 update from the Georgetown University Center on Education and the Workforce reports that nearly 28% of associate degree holders now earn more than the median bachelor's degree recipient, especially within high-demand technical fields such as health sciences and engineering technology (Carnevale, Cheah, & Wenzinger, 2021). This challenges the assumption that more education always equates to higher earnings and suggests that the field of study and job-market alignment may be better predictors of success than the degree itself.

Further building on this, Johnson (2024) provides a detailed analysis of the cost-effectiveness of trade schools and bootcamps, showing that many of these programs lead to similar or even higher job placement rates compared to four-year colleges. Not only do they help students avoid excessive debt, but alternative programs also often help students enter the workforce quickly and efficiently by focusing on targeted skills and offering shorter time frames. Focusing specifically on skill development, alternative programs often deliver a better ROI as well, particularly for students who prefer quick transitions into stable employment.

As opportunities for skill-building expand, so do employer expectations. According to Deloitte's 2022 research, 68% of business leaders no longer view a bachelor's degree as a requirement for many roles, and 85% consider alternative credentials such as boot camps, certificates, or online courses legitimate qualifications (Deloitte, 2022). This marks a significant shift from prior decades when degrees functioned as a universal screening tool. In industries like technology and digital media, where practical ability is highly valued, demonstrable skills are increasingly prioritized over formal credentials. Employers now place greater emphasis on candidates' ability to showcase their capabilities through project portfolios and real-world experience, rather than relying solely on educational background.

Another significant development is the shift in the opinion of online learning as a credible educational path. Athey and Palikot's (2024) randomized study, involving over 800,000 Coursera users, found that learners who publicly shared their massive open online course(MOOC) credentials on platforms like LinkedIn significantly improved their employment outcomes. For those with a low baseline employability (as calculated in the study), this simple act more than doubled their chances of securing relevant work. This shows the rising influence of job-specific skill building and sharing, where credentials are not just earned but can also be shared and verified instantly. This research confirms that online platforms are no longer supplementary. Their flexibility, affordability, and ability to connect learners with increasingly recognized credentials position them as huge contributors to workforce readiness, particularly for nontraditional students.

Unfortunately, structural challenges remain, as evidenced by Brozina and Johri (2024), who highlight an issue in educational policy: the lack of a clear and consistent definition of “nontraditional student.” Their analysis reveals that the ambiguous terminology leads to gaps in institutional support, making it extremely difficult to create effective programs and outreach for those pursuing alternative paths. Without clarity, students fall through the cracks, unfortunately putting those balancing external circumstances at a higher disadvantage.

Bain & Company’s (2022) report offers a solution in the form of career-connected learning. By integrating real-world work experiences, internships, and employer partnerships into the education pipeline, students are more engaged and better prepared for employment. These programs not only improve job placement rates but also increase motivation and satisfaction by showing students the direct relevance of their learning. Bain emphasizes that scaling these models is critical for making alternative pathways both accessible and sustainable.

Today’s research doesn’t just validate the concerns raised before 2020; it amplifies them. The education-to-career route is no longer built on prestige or tradition, but on proof and hard work: demonstrated skills, measurable outcomes, and the ability to adapt in real time. As employers pivot toward competency- and skill-based hiring, students are opting for faster, more flexible pathways that align with their ambitions, resources, and goals. In this continuously evolving setting, success favors those who will build, apply, and signal their value, not just earn it on paper.

IV. What’s Changed

With just how many different things that COVID-19 changed our world, none seemed to truly compare with how it affected our workforce. Never before did we imagine our world would essentially be put on halt, and practically fully virtual, and yet the pandemic made it a reality. And with such a a controlling affect on the workforce, arguably the most glaring of the issues it caused within it was how it highlighted

the problems within the current higher educational model. For millions of people, their previously thought to be extremely safe and reliable investment of their four-year degree turned out to not be able to protect them from the economic turmoil the pandemic brought and made us question their true worth. Leading up to the pandemic, there were already many perceptions and studies beginning to shift this outlook, and it turned out that the pandemic only proved them right and pushed this massive shift of alternative pathways to careers forward.

This disruption pushed not only students but also employers to question the actual return on investment of a degree with more pressure than ever before. The previous notion of degrees being a requirement for most careers led to a strong push to uncover the true ROI. Suddenly, we observed a growing number of industries quickly adapting to this change, shifting their hiring requirements to emphasize skills proven and projects completed, a significant departure from the traditional view of a degree as the ultimate requirement. As Deloitte's 2022 report highlights, companies across industries began to prioritize competency-based hiring practices, moving away from rigid degree requirements. They realized that the degrees alone were a poor indicator to purely judge whether someone was the right fit for the role and started to prioritize skills and certifications to access a broader, and hopefully better, talent pool. With this shift in employers priorities, we saw that students reacted efficiently. Increasingly, students exploited those alternative pathways, such as bootcamps, online learning programs, certificate programs, and more. And fascinatingly enough, in many cases, those with targeted, practical skills found themselves at an advantage over peers with more generalized four-year degrees but less demonstrable experience.

With this radical change, the question of what truly is the "value" that education is trying to serve for both the students and employers still remains. Is the knowledge gained from education what provides the value, or are the skills that are applied into experience what truly are more rewarding? Which has the better return on investment? To best answer this question, we must understand that research suggests it truly depends. For many people with learning styles that works best for them, or for many professions,

where experience truly does outweigh the knowledge, alternative pathways are the answer and have a much better ROI.

To best answer this question, you must understand there is not a direct answer. Similar to many answers to hard questions within human capital, the answer is that it depends. Seemingly endless factors come into play to determine what the path to a given career is, whether that is the macro level of the skills that the role requires all the way down to the specific individual and which learning method is most optimal for them personally. The key here is that research and the future of work suggest that there is now an evolving growth of those paths, which means that there will be more ways to both achieve a given career and more ways for a company to find that perfect fit. So in the end, whether the knowledge or experience is more valuable depends on a multitude of factors ranging from the role to the individual and their own circumstances.

Ultimately, this phase of our research reveals just how pivotal of a moment in the history of human capital. The pandemic accelerated trends that were already in development, and the resulting questions and insights are reshaping the entire narrative around success after high school.

V. Future Directions

The rise of alternative pathways is not a threat to higher education; it is an opportunity to serve students better. Traditional college degrees still hold significant value, but only when chosen with the right goals, fields, and cost-benefit analysis in mind. As research has shown, blind allegiance to traditional secondary education no longer guarantees a strong return on investment or long-term career success. Instead, the best path today is one aligned with individual ambition, market demand, and practical skill-building. Students deserve the freedom and the guidance to pursue the route that fits their needs, not outdated societal expectations.

To fully realize this opportunity, systemic changes must follow. Schools must expand support for modular and stackable credentials that allow students to build flexible, market-responsive educational experiences. Policymakers must focus on improving transparency around career outcomes, wages, and employment stability across all types of education, not just four-year degrees. Future research should prioritize developing a clear, modern definition of "nontraditional students," as Brozina and Johri (2024) identified, to ensure that policies and programs reflect today's learners. Future studies tracking career outcomes by pathway, audits of institutional barriers to credit transfer, and surveys of evolving employer preferences are also urgently needed. Without this research, students will continue to face information gaps that leave them vulnerable to poor investment decisions. Strengthening these areas will not only support individual success but also help create a more equitable and resilient education system overall.

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